SW2000TSN SWITCH

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## 100/1000BASE-T1 SWITCH

**SW2000TSN** SWITCH

**FEATURE**:

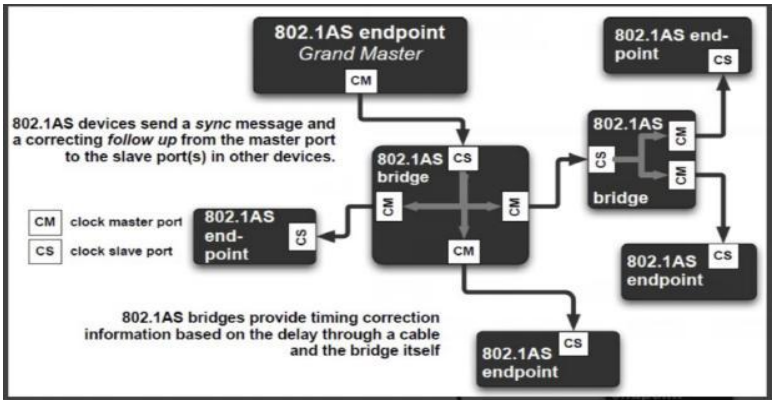
- 8ch 100/1000Base-T1
- 1ch 1/2.5/10GBase-T SFP
- 2ch 100/1000base-T RJ45
- 802.1AS-2020 gPTP (BC)
- 802.1Qbv/ci TSN
- 802.1Q VLAN
- AVB support
- DIP config mode/speed
- Metal Shell and Status LED
- H-MTD and MATEnet
- With RJ45/SFP+

# SW2000 TSN SWITCH
# USER MANUAL
# 100/1000BASE-T1SWITCH

**SW2000 TSN Switch**,

11 port Gigabit/100Mbps Automotive Ethernet TSN time sensitive switch, which includes 8 channels of 100/1000BASE-T1 using Rosenborg H-MTD or Tyco MATEnet interface, 1 channel 1/2.5/5/10G standard Ethernet SFP and 2 channels 1000BASE-T standard Ethernet RJ45, it can achieve multi-channel switching of Automotive Ethernet, Gateway, AVB application, gPTP boundary clock (BC), gPTP clock bridge, Bypass data monitoring and other functions. It can also be used for communication data collection and analysis between multiple onboard ECUs

**System architecture:**




**Feature:**

- Power supply: 7V~17V， temperature:-40℃ to +85℃
- Widely used in the fuel and EV TSN system, lidar/radar, camera, ADAS. Industrial control and aerospace。
- The static current power consumption is about 6W, and the full load current power consumption is about 7.5W.
- The Master and Slave for each port, as well as the communication rate configuration, are set by the DIP switch.
- It has overvoltage protection, reverse connection prevention, surge impact prevention, and is suitable for long-term vehicle operation.
- Size: 234x 114 x 32mm (SW2000TSN Gen 2 Support 24V/48V.)

# DOCUMENT REVISION HISTORY

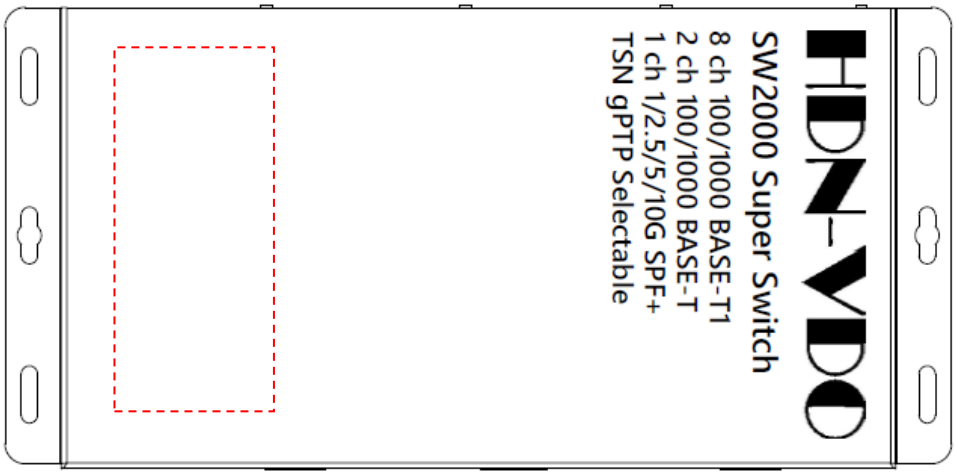
| Revision | Modification time | Revise contents |
|---|---|---|
| Rev1 | 2023-07-08 | Initial version |
| Rev2 | 2023-12-10 | Increase the usage of configuration tools and the application of PTP |
| Rev3 | 2024-06-17 | Adjust the technical parameters |
| Rev4 | 2024-09-07 | Add and revise technical solutions, apply references, and conduct engineering applications |
| Rev5 | 2025-01-25 | Increase the dimensions of the shape and the installation size |
| Rev6 | 2025-05-29 | Add configuration items such as 802.1as, Qbv, VLAN, Membership, etc. |
| Rev7 | 2025-09-29 | Add power interface and Ethernet interface models, definitions, and wiring reference |
| Rev8 | 2026-01-19 | Update the document, add MAC address binding for ATU configuration, data frame filtering and processing using TCAM configuration. The second generation can support higher voltages. It can be applied to 24V and 48V systems. |

# OPERATING

The SW2000 series Automotive Ethernet switch has different configurations, which enable it to provide customers with powerful application functions and meet their more precise application needs. Specifically, it is divided into four configuration options:

| Product Model | SW2000M TSN | SW2000H TSN | SW2000M | SW2000H |
|---|---|---|---|---|
| Connector type | MATEnet | H-MTD | MATEnet | H-MTD |
| TSN support | **Yes** | **Yes** | No | No |
| Others | **Yes** | **Yes** | **Yes** | **Yes** |

The top view of the product is as follows. A label is pasted at the position of the red dashed box to display the product model.

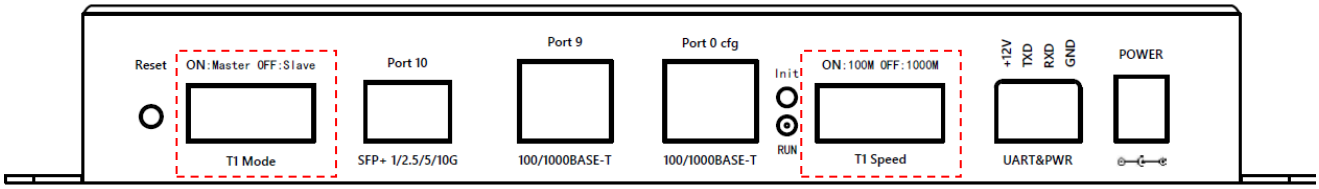



MATEnet connector version:

H-MTD connector version:

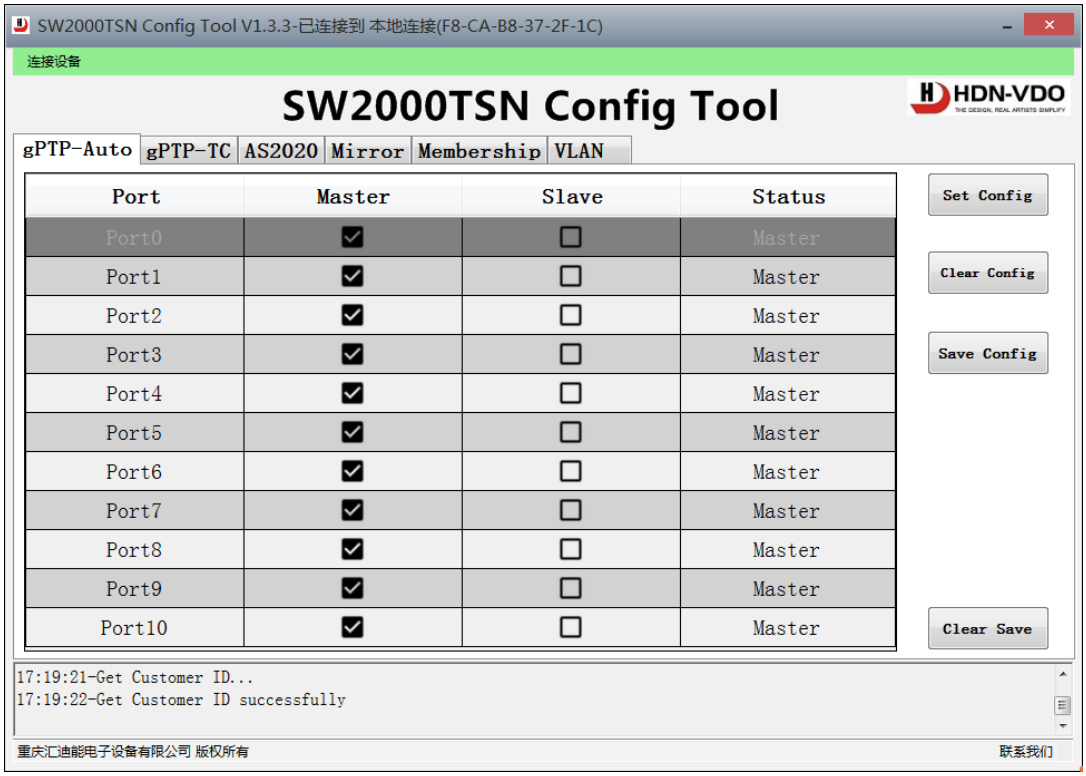
The product uses a T1 Mode dip switch to configure 8 channels of 100/1000BASE-T1 channels for Master and Slave. The internal control unit sets and activates the master-slave relationship of each port by reading the configuration results of DIP dip switches.

At the same time, the product uses a T1 Speed dip switch to configure whether the speed of the 8 channels is 100BASE-T1 or 1000BASE-T1. The following is the layout of the device interface and the configuration position of the dip switch:




Note: After changing the configuration of the T1 Mode and T1 Speed dip switches, the function can only take effect after pressing the Reset button or a power-off cycle.

SW2000TSN Config Tool:



# SW2000M/H TSN series power interface definition

## Molex interface (involving models: SW2000M TSN, SW2000H TSN)

**Device end:**

| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Molex mini50 Base | 34793-0040 | 1- +12V<br>2- NC-Not connect<br>3- NC-Not connect<br>4- GND | Molex |

**Wire harness end：**

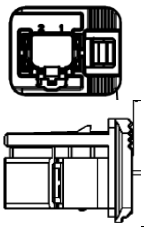
| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Molex mini50 Plug | 34791-0040 | 1- +12V<br>2- NC-Not connect<br>3- NC-Not connect<br>4- GND | Molex |
| Cable Core | 43030-0001 | - | Molex |

# SW2000M TSN series harness interface definition

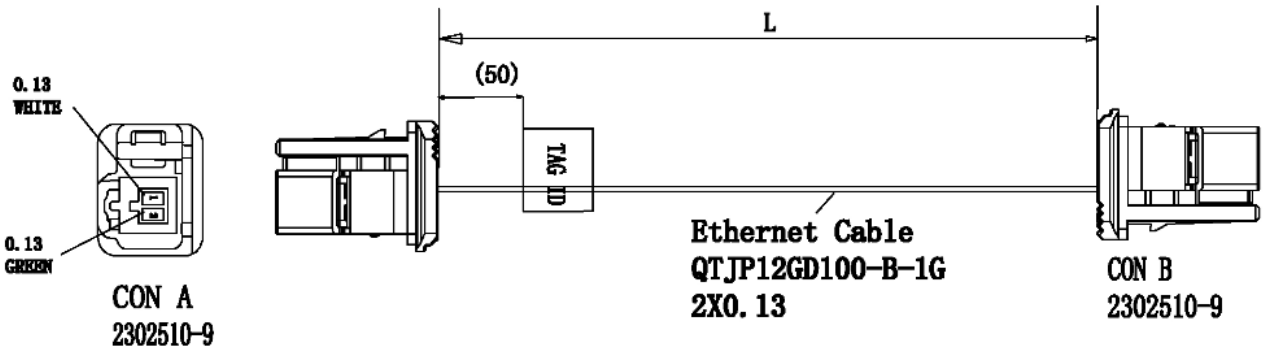
**Device end:**

| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Tyco MATEnet Port 1 P M | 2304372-1 | 1- N<br>2- P | Tyco |

**Wire harness end：**

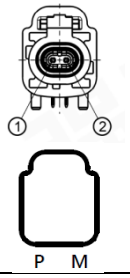
| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Tyco MATEnet  | 2302510-9 | 1- N<br>2- P | Tyco |

参考线束：

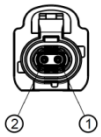
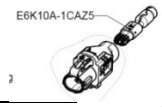


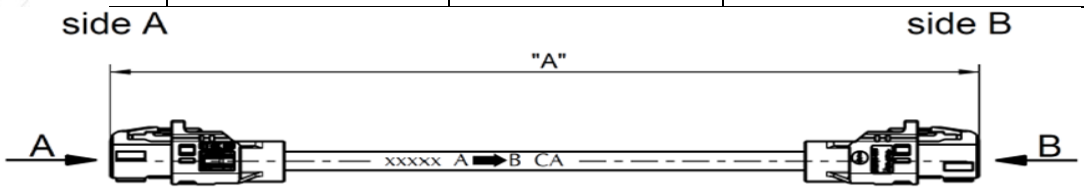
# SW2000H TSN series harness interface definition

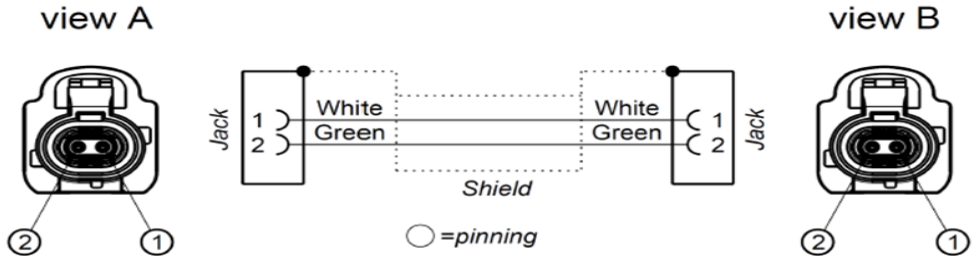
**Device end:**

| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Rosenberger H-MTD   | E6S20A-40MT5-Y | 1- P<br>2- N | Rosenberger |

**Wire harness end：**

| Name | Model | definition | Manufacturer |
|---|---|---|---|
| Rosenberger H-MTD   | LCA-101-XXX-Y-Y | 1- P<br>2- N | Rosenberger |
| Core   | E6K10-1CAZ5 | 1- P<br>2- N | Rosenberger |




| Dimensions (mm) | | | |
|---|---|---|---|
| Range (length "A") | 200 to 1000 | 1001 to 5000 | 5001 to 20000 |
| Tolerance (mm) | ± 10 | ± 15 | ± 20 |

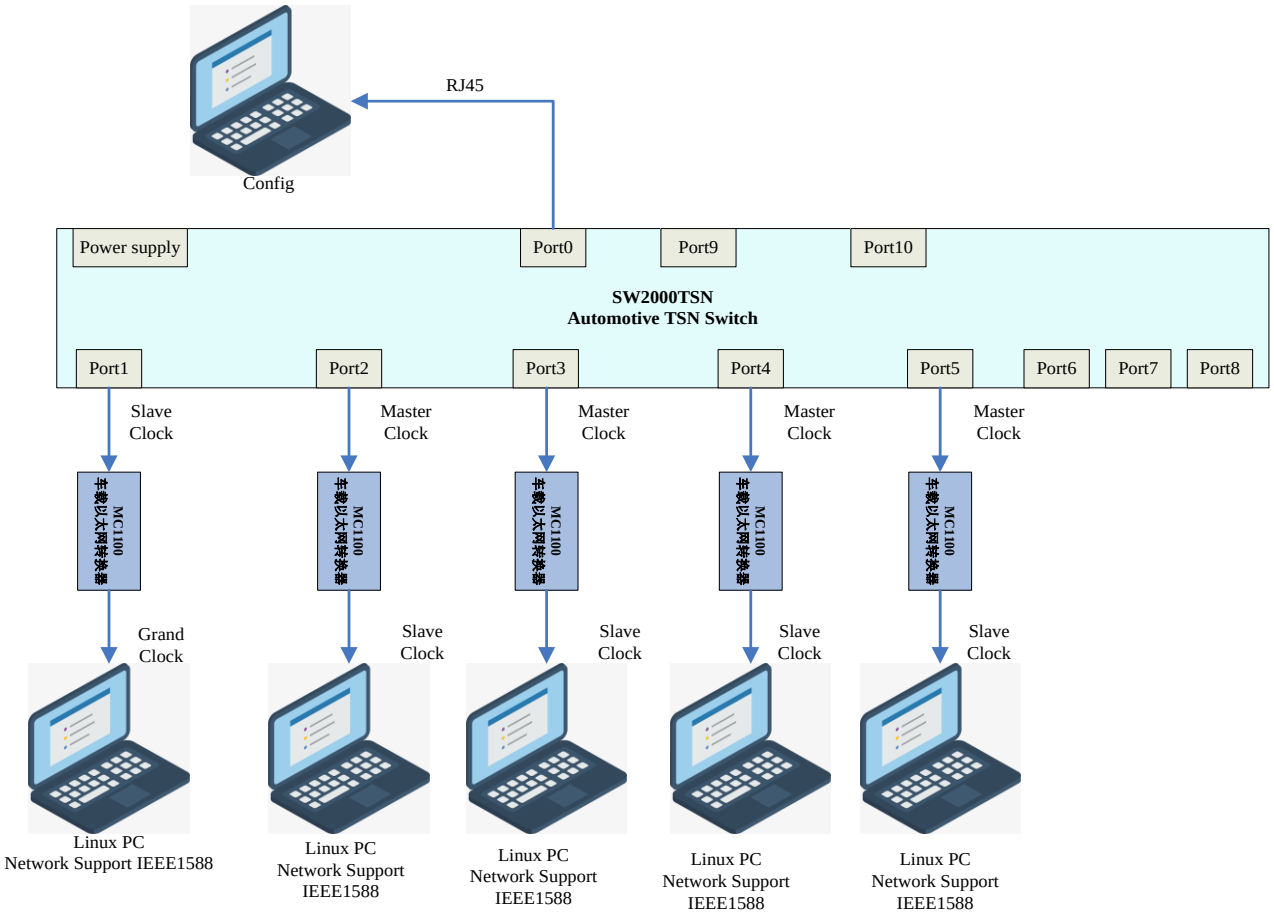
# 2 SOLUTIONS

## 2.1 Dual channel as a monitoring and data collector (Bypass function)

This product can capture communication messages between DUT1 and DUT2, and can also simulate nodes to send messages to DUT1 and DUT2. At the same time, data exchange between multiple master devices and multiple slave devices can also be achieved. Communication data of each master node can be viewed through monitoring software.

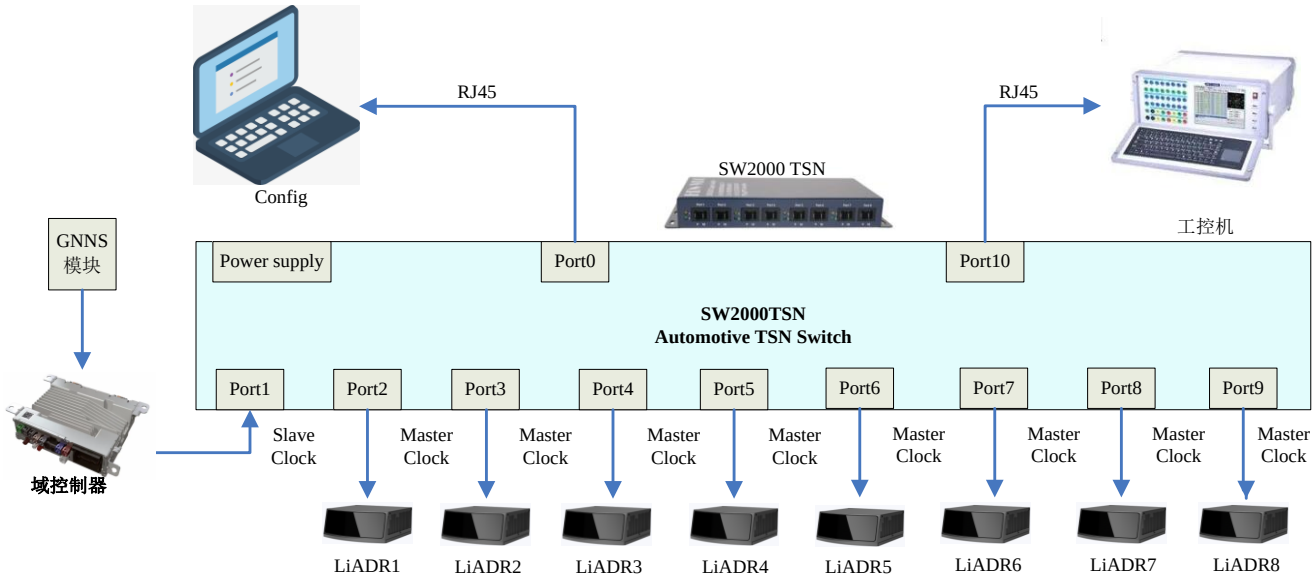
## 2.2 Case of building gPTP clock bridge experiment

The Automotive Ethernet TSN test network aims to build the necessary network architecture for precise time synchronization and simulation of vehicle internal controller modules for accurate network timing. It requires one grand clock node, one or more slave clock nodes, one onboard Ethernet TSN switch, and one configured computer. Among them, the ancestor clock and slave clock nodes are simulated and implemented using Linux computers (network cards must support IEEE1588) and vehicle mounted Ethernet converters. Of course, each computer and converter combination can be replaced with actual LiDAR or other slave clock nodes. The system architecture is as follows:



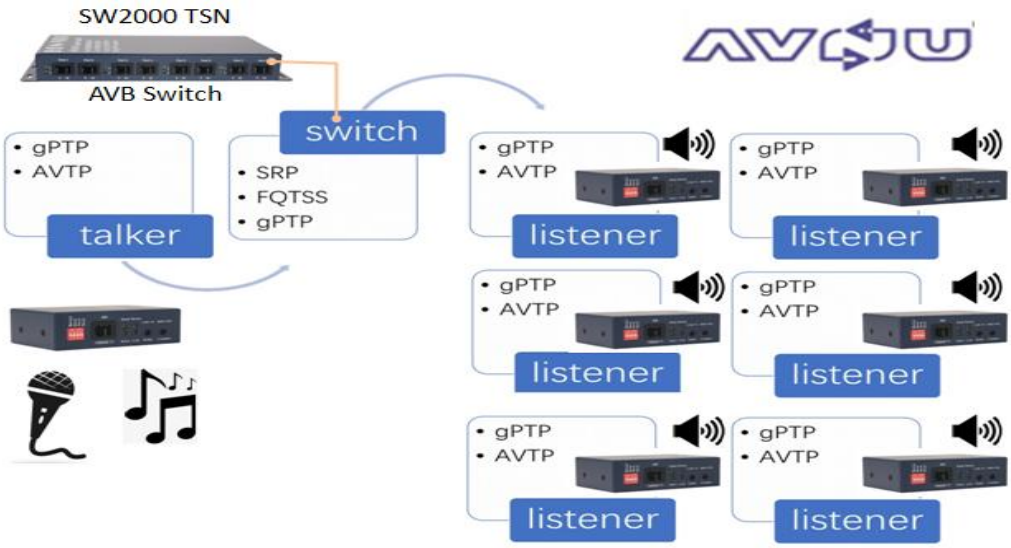
2.2 Case of gPTP clock bridge for automotive LiDAR

Build a simplified laser radar bridging scheme as follows, supporting gPTP time synchronization and achieving data transmission and precise synchronization between the main control module and the laser radar. Other similar applications can also refer to this solution, such as high-definition cameras in vehicles, millimeter wave radar, industrial robots, avionics system clock synchronization, and so on.



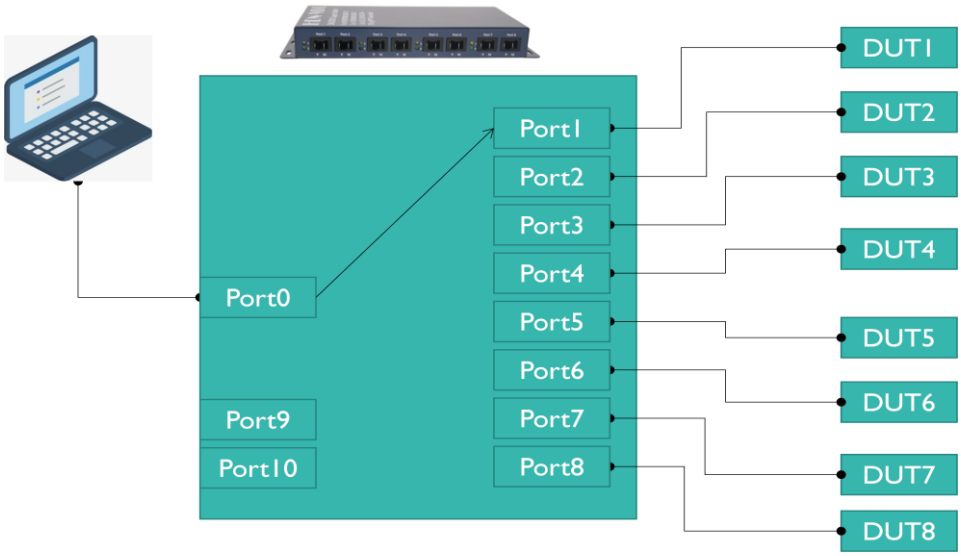

2.3 AVB Audio Video Bridge Application Solution

The following is a simplified AVB audio and video bridge application solution that supports gPTP time synchronization. Other applications with similar solutions can also refer to this solution, such as AVB node networking, audio and sound layout, industrial high-definition cameras, millimeter wave radar, industrial audio and video synchronization system, long-distance audio and video synchronization system over 10km, and so on.




2.4 Automated testing production line solution

When building an automatic testing production line, it is necessary for the industrial control computer to communicate with multiple targets one by one through industrial Ethernet RJ45, but the number of Ethernet interfaces for the industrial control computer is limited. Moreover, using a regular switch cannot locate or specify independent data interaction to a specific port. SW2000 provides a means of channel isolation by sending Ethernet message instructions through an industrial computer to set up SW2000, allowing a certain channel, such as Port0, to communicate separately with any other channel, making operation convenient and switching efficient.



# 3 BRIEF INSTRUCTIONS FOR USE

Connect the TRX-N and TRX-P wires of one channel of the SW2000TSN Ethernet switch to the corresponding test target, and configure the port mode as Master or Slave as needed If the target device is a Master, set the port to Slave, and vice versa. Add 12V power to the switch device again. At this time, the yellow Init light and green RUN light of the switch will light up. When the yellow Init light goes out, it means that the switch initialization is complete and can be used. At this point, use a regular Ethernet cable to connect the computer to the Port0/config port of the SW2000TSN switch.

If the target device is correctly connected to the port and the device and corresponding channel are configured as Master and Slave respectively, that is, the matching is correct, the indicator light of the corresponding channel will light up and flash. At the same time, it also indicates that the channel network has established a data link. This way, the Ethernet software on the computer can be used to monitor and collect data from the tested target. At the same time, SW2000TSN Config Tool can also be used for configuring specific functions.

When the indicator light is normal, the Ethernet software on the computer can be used to monitor and collect data from the tested target.

Definition and Explanation of LED Lights:

| LED name | color | function |
|---|---|---|
| Init | yellow | Light up after startup or restart, and turn off after initialization is complete. |
| RUN | green | Solid green after being powered on. |
| T1 channel LED | yellow /green | Connect to the target device, light up after successful speed and master-slave matching. Flashing when there is data communication. |

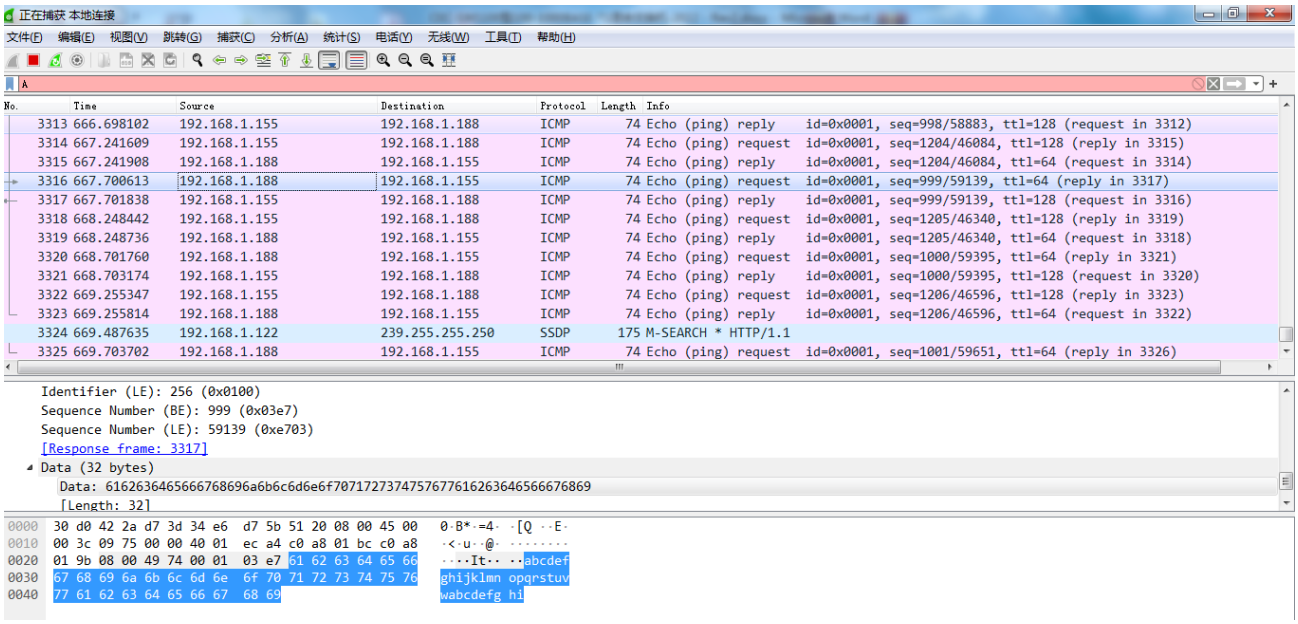
For example, the IP address of the tested device is 192.168.1.122 Configure the computer's IP address to 192.168.1.155, which is in the same LAN segment. In this way, you can open the DOS terminal on your computer and use the Ping 192.168.1.122 -t command to send data packets to the target device.

Test case (taking 1000BASE-T1 port as an example, 100BASE-T1 port is similar):

The tested device DUT1 is the Master of 1000BASE-T1 (IP: 192.168.1.122), and the tested device DUT2 is the Slave of 1000BASE-T1 (IP: 192.168.1.188). Connect the two devices to CH1 and CH2 of SW2000TSN respectively Among them, CH1 is configured to connect Slave and DUT1 channels, CH2 configures this channel to connect Maser and DUT2 After connecting the power supply, The indicator lights corresponding to channels CH1 and CH2 are lit, indicating that a connection has been established and the indicator lights will flash when there is data communication

Then, connect the SW2000TSN to the computer using an Ethernet cable and configure the computer's IP to be on the same network segment as the two devices (IP: 192.168.1.155), Open Wireshark software to view the communication data between DUT1 and DUT2

WiresharkDisplay as follows:




Of course, you can also use the data collection software and Ethernet transceiver testing tools we provide:

Wireshark Tool:

https://wwm.lanzouh.com/iK3eQtjvfti Password:d5te

Data send tool:

https://wwm.lanzouh.com/i3bNTtjv7hi Password:4pd7

Testing tool:

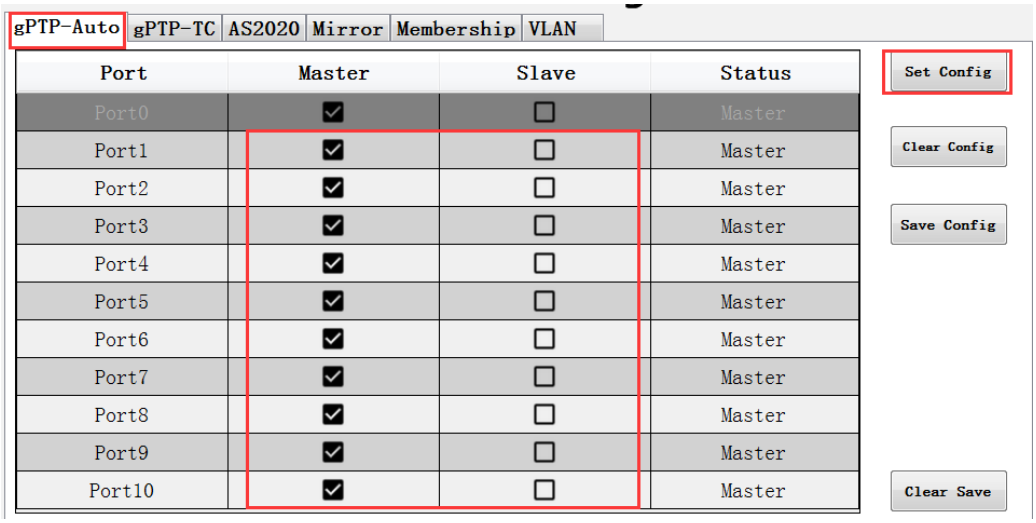
https://wwm.lanzouo.com/b01psw5vi Password:bn3z

SW2000TSN Config Tool please reach out to technical team

# 4. CLOCK SYNCHRONIZATION GPTP CONFIGURATION INSTRUCTIONS

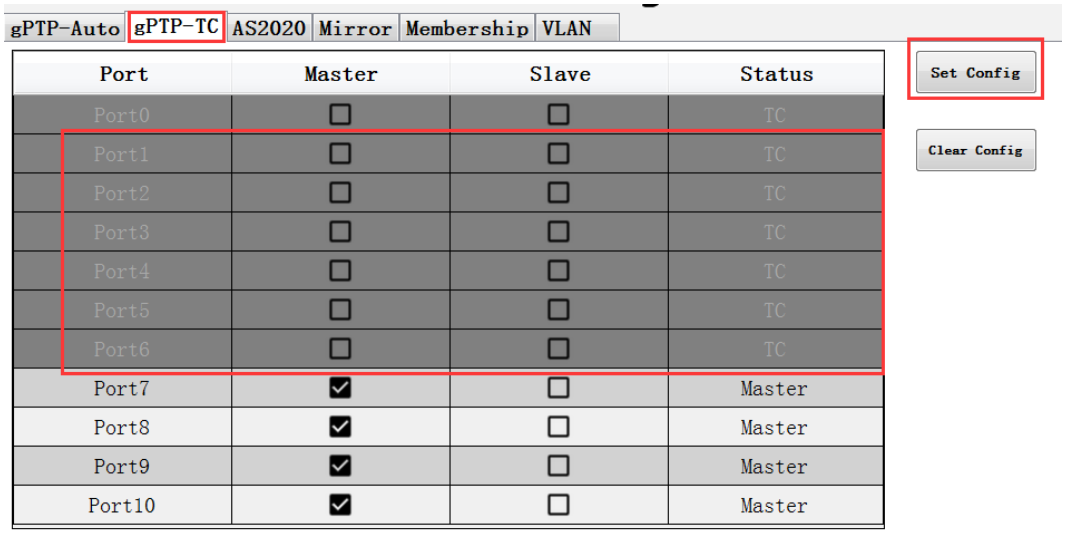
## 4.1 Configuration 1 -- gPTP Automotive

This device supports gPTP configuration based on time synchronization within the automotive domain. You can open the configuration software, select the gPTP-Auto configuration tab, and set the clocks for each channel to meet the system requirements.



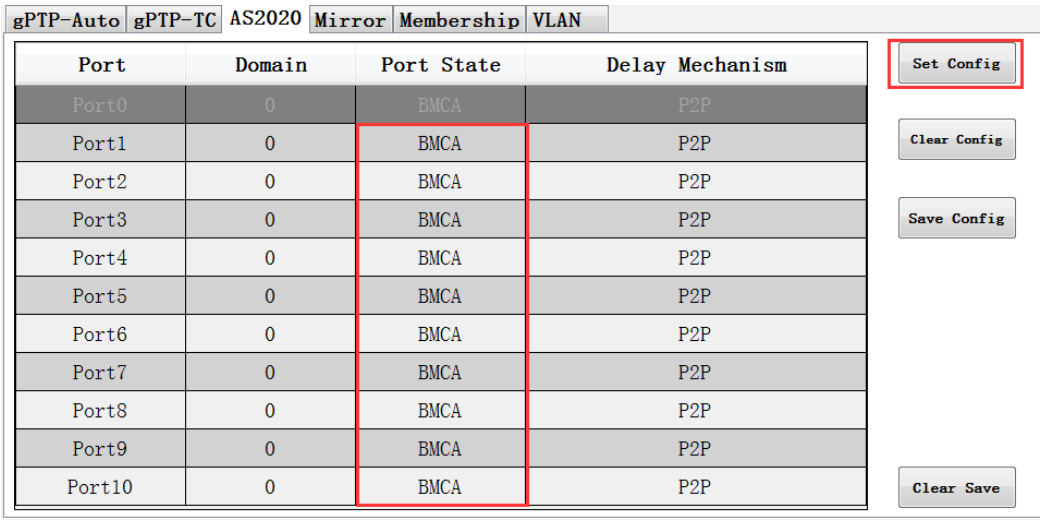

## 4.2 Configuration 2 -- gPTP TC

This device supports acting as a TC transparent clock to achieve direct forwarding of time messages. Select the gPTP-TC configuration label.



### 4.3 Configuration 3 - AS2020

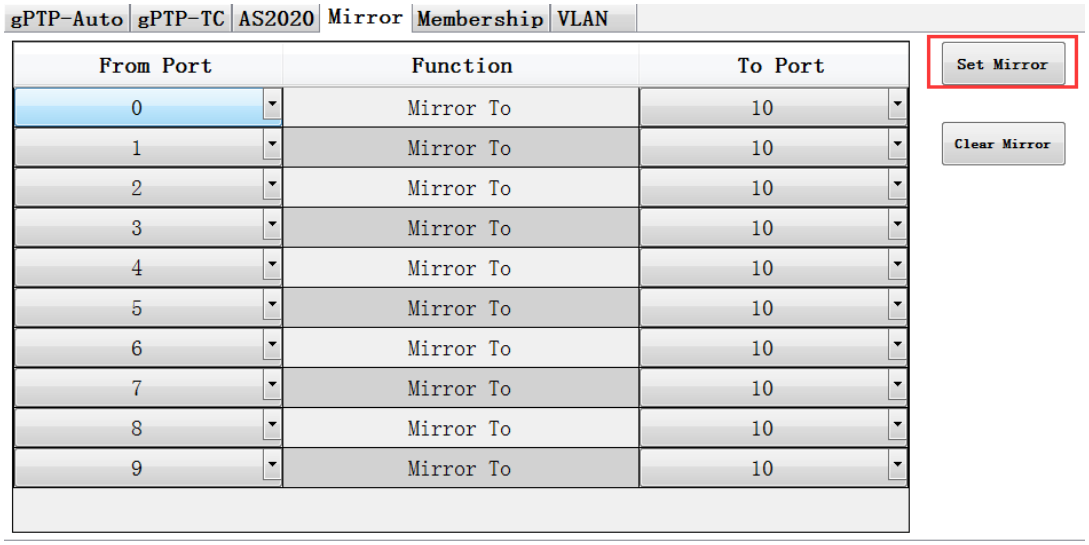
This device supports gPTP-based time synchronization configuration within the vehicle domain. You can open the configuration software and select the AS2020 configuration tab. This function supports the BMCA algorithm and is a typical application of gPTP.




| Port | Domain | Port State | Delay Mechanism |
|---|---|---|---|
| Port0 | 0 | BMCA | P2P |
| Port1 | 0 | BMCA | P2P |
| Port2 | 0 | BMCA | P2P |
| Port3 | 0 | BMCA | P2P |
| Port4 | 0 | BMCA | P2P |
| Port5 | 0 | BMCA | P2P |
| Port6 | 0 | BMCA | P2P |
| Port7 | 0 | BMCA | P2P |
| Port8 | 0 | BMCA | P2P |
| Port9 | 0 | BMCA | P2P |
| Port10 | 0 | BMCA | P2P |

# 5 PORT MIRRORING CONFIGURATION INSTRUCTIONS

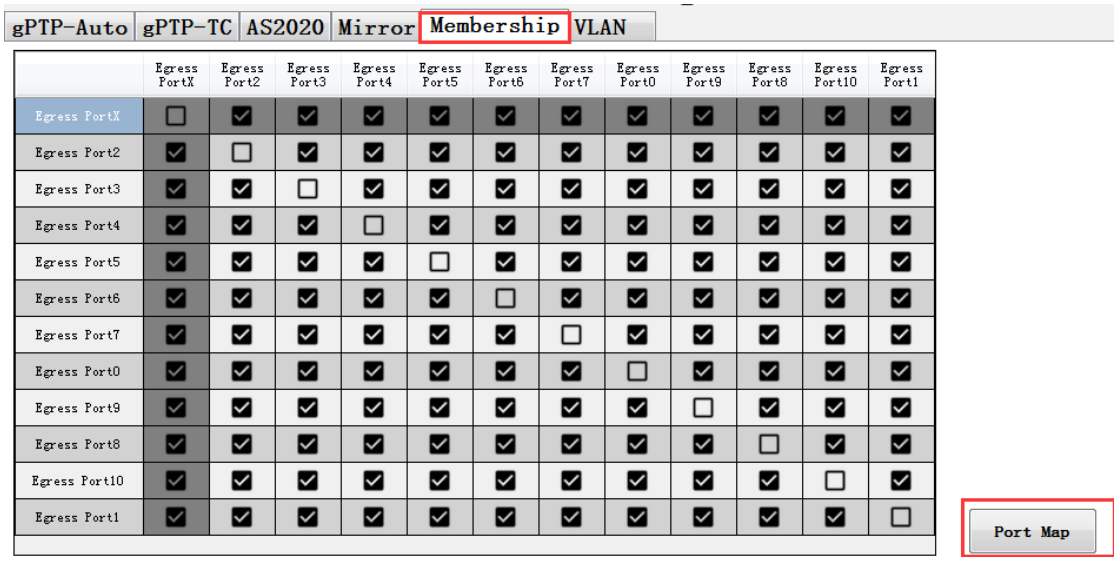
This device supports packet port mirroring, allowing the mirroring of traffic from other ports to a designated port. This feature is suitable for data analysis applications. All the ports that need to be observed are mapped to the designated port, connected to a computer, and analyzed using Wireshark. (Note: The mirrored ports do not support gPTP synchronization.)




| From Port | Function | To Port |
|---|---|---|
| 0 | Mirror To | 10 |
| 1 | Mirror To | 10 |
| 2 | Mirror To | 10 |
| 3 | Mirror To | 10 |
| 4 | Mirror To | 10 |
| 5 | Mirror To | 10 |
| 6 | Mirror To | 10 |
| 7 | Mirror To | 10 |
| 8 | Mirror To | 10 |
| 9 | Mirror To | 10 |

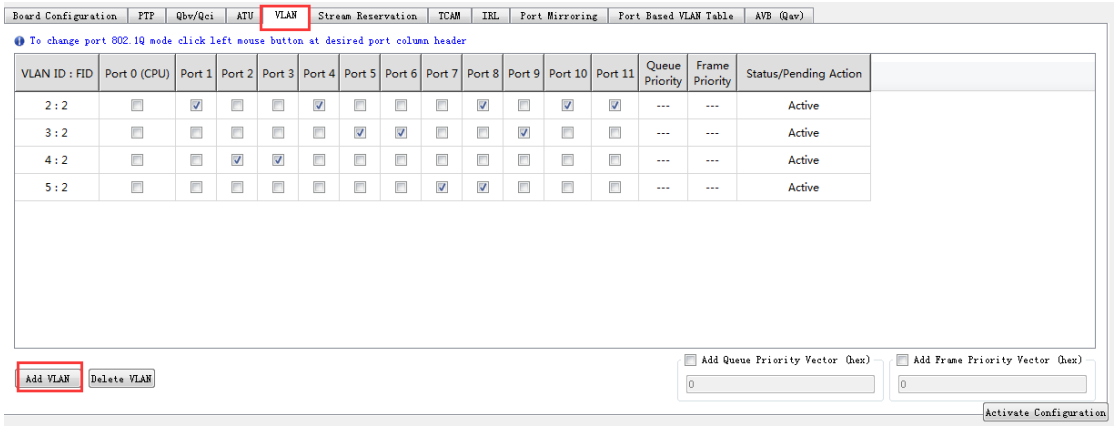
# 6 PORT MEMBERSHIP CONFIGURATION DESCRIPTION

This device supports port grouping, enabling packet distribution grouping and isolation. This function is suitable for data packet forwarding and isolation of ports.



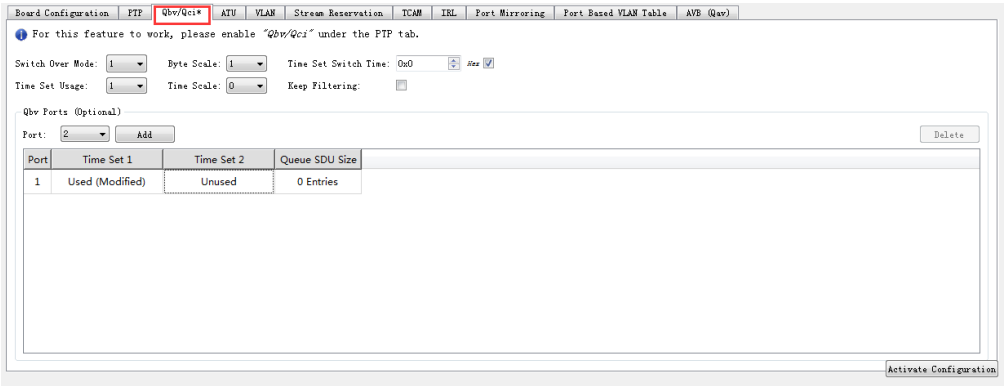

# 7 802.1Q VLAN CONFIGURATION INSTRUCTIONS

This device supports the 802.1Q protocol and allows users to arbitrarily divide VLANs to implement data forwarding policies between ports.



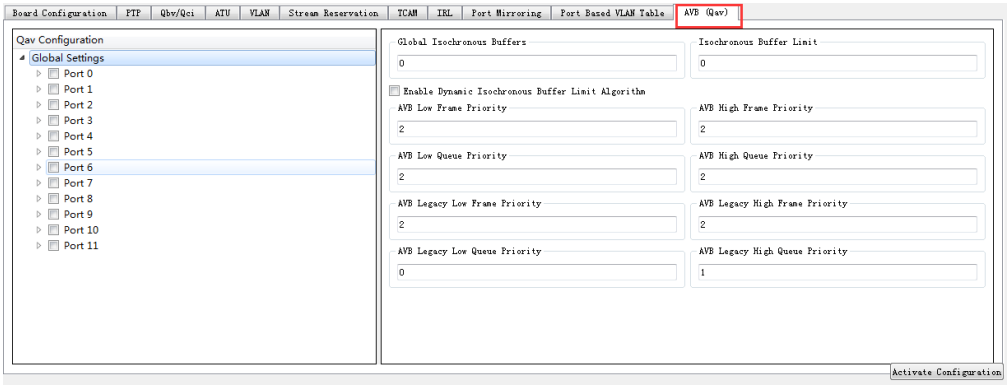

# 8 TSN 802.1QBV/CI CONFIGURATION INSTRUCTIONS

This device supports the relevant functions of the 802.1Qbv/ci protocol in the TSN protocol. It supports the configuration of various option parameters through the interface. (Note: Functions 8-10 are suitable for engineers with some knowledge of the protocol. We can provide technical support such as software usage.)



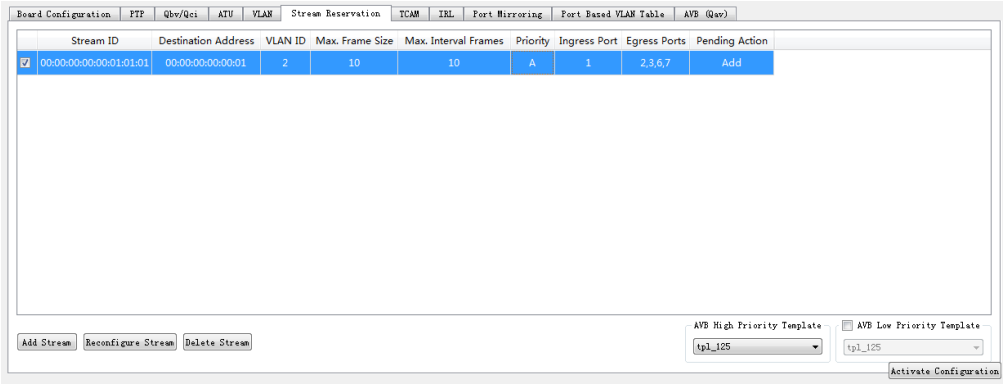

# 9 AVB 802.1QAV CONFIGURATION INSTRUCTIONS

This device supports the relevant functions of the AVB protocol 802.1Qav and allows configuration of various option parameters through the interface.



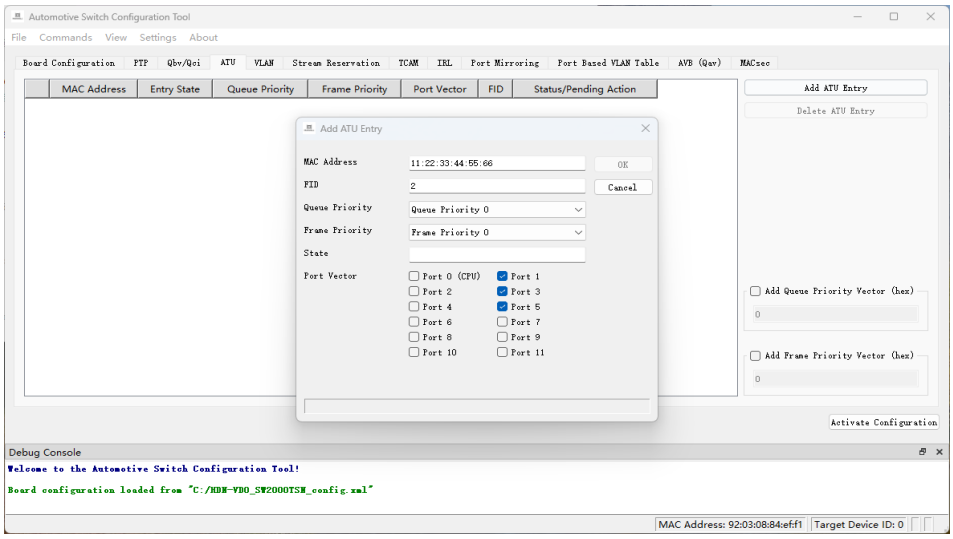

# 10 STREAM RESERVATION CONFIGURATION GUIDE

This device supports the related functions of Steam Reservation and allows configuration of various option parameters through the interface.



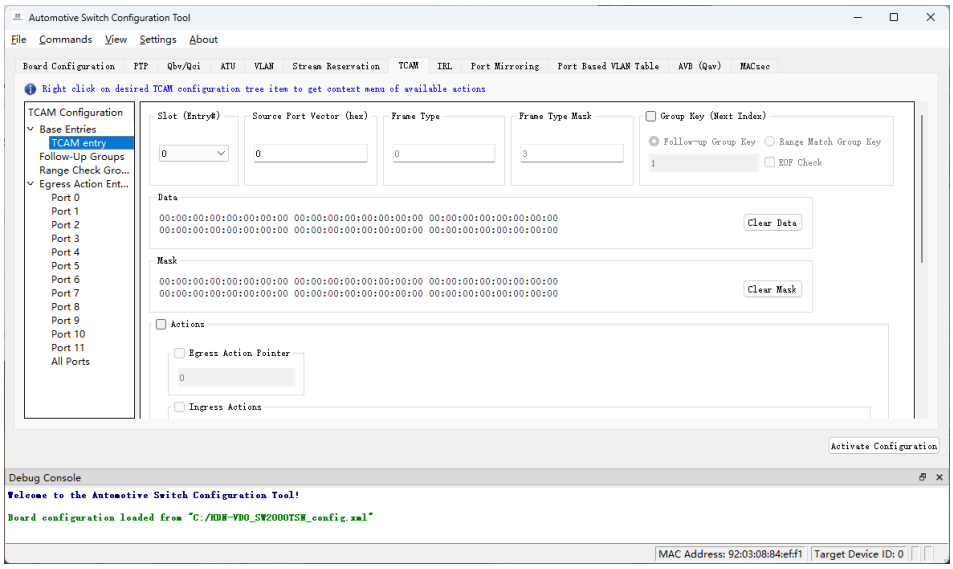
# 11 CUSTOMIZED BINDING CONFIGURATION INSTRUCTIONS FOR ATU OF MAC

This device supports the function of port MAC address binding and allows configuration through the interface.




# 12 ETHERNET DATA FILTERING RULES AND PROCESSING CONFIGURATION INSTRUCTIONS

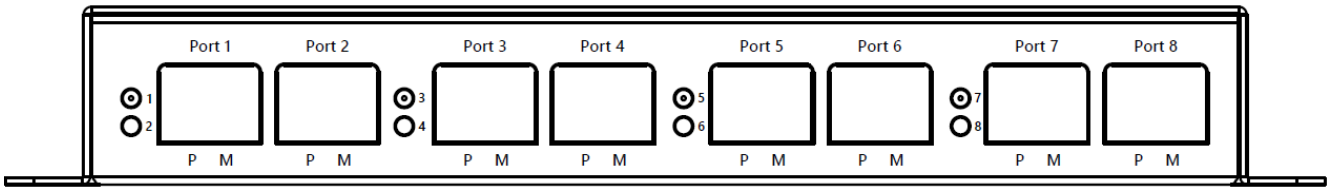
This device supports port data filtering and processing functions such as modification/discard, etc. It also supports configuration through the interface.



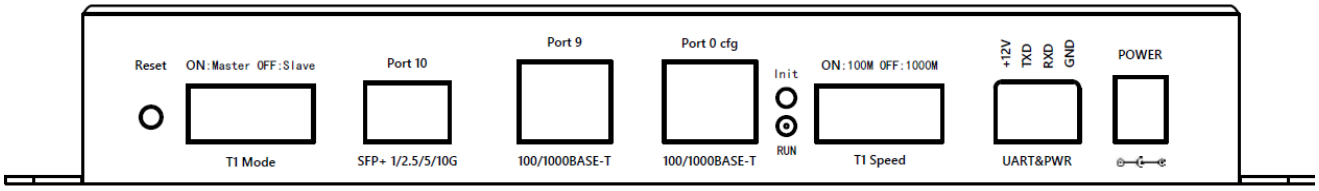
# Attach:

## SW2000TSN Size

Front side




Back side

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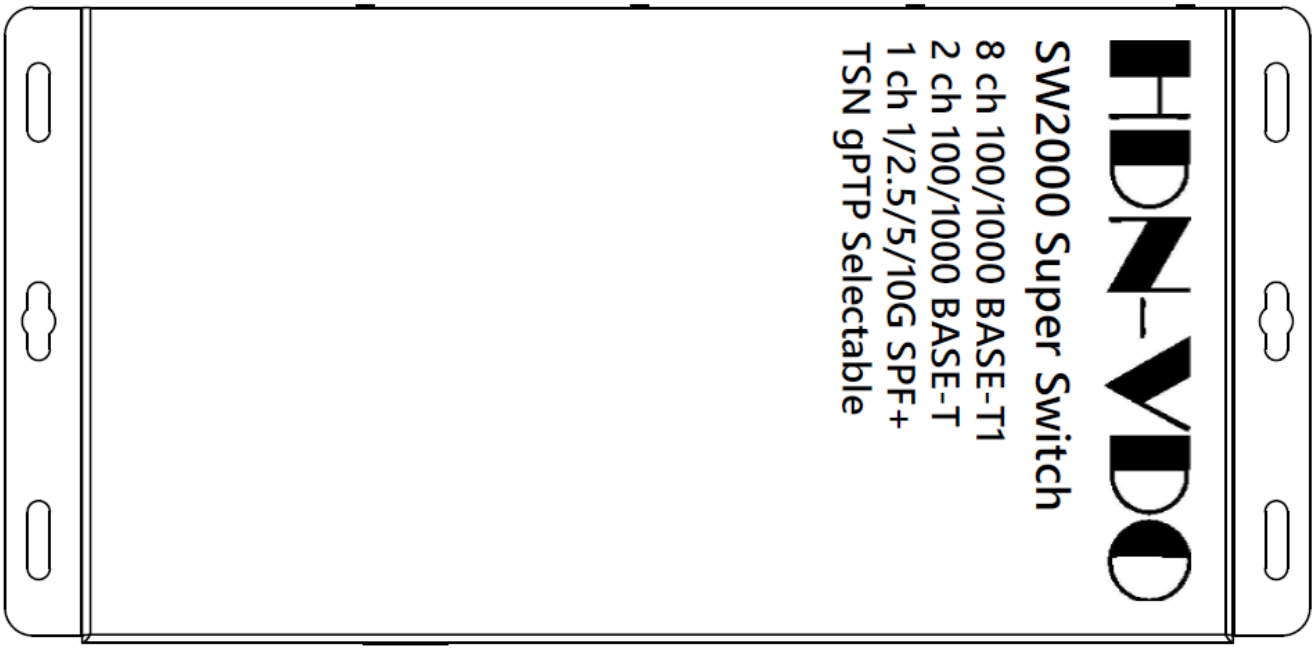
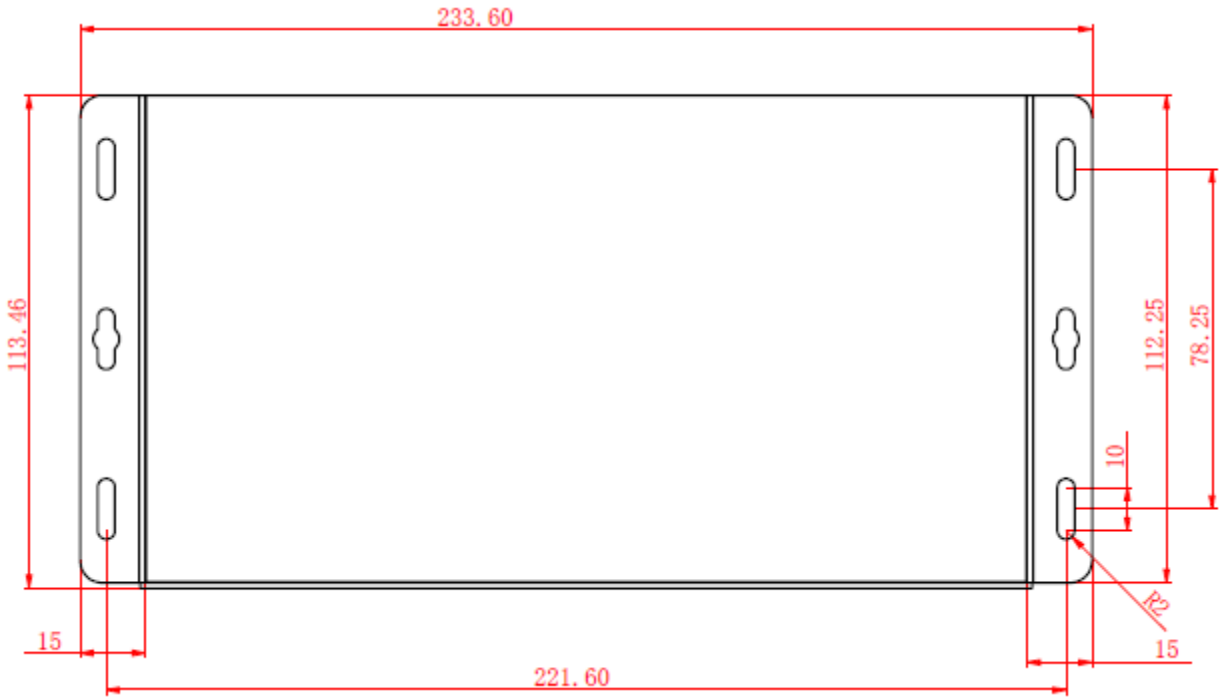
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